DISCUSSION

A NEW ARCHAEOLOGICAL FIND IN THE GULF OF CAMBAY, GUJARAT

by S. Kathiroli, S. Badrinarayanan, D. Venkata Rao, S.N. Rajaguru, K.M. Sivakholundu and B. Sasisekaran. Jour. Geol. Soc. India, v.60, pp.419-428

R.V. Karanth, Department of Geology, M.S. University of Baroda, Vadodara - 390 002, comments:

The authors deserve to be congratulated for the discovery of the new archaeological find in the Gulf of Cambay, Gujarat (Kathiroli et al. 2002). However, there are a few aspects which need to be clarified:

1. It is improper to correlate the E-W alignment of palaeochannels with the direction of course of Tapi river on the eastern side of the Gulf with that of the Satrunji river on the west. If the course of Tapi river is taken as the continuation of Satrunji river channel in the west, Kalubhar river could likely be the western extension of Narmada river and Ghelo Nadi/Keri Nadi be taken as western extension of Mahi river and so on. Is it desirable to take such a correlation of E-W extension of river systems? Whether the river systems on either side of a rift are correlatable with such ease?
2. The broken barrel shaped bead (Fig.10) measures 13 mm in length, 6 mm in width with a 4 mm hole, which leaves its wall to be as thin as 1 mm. It is amazing that such a bead was made from a material like chert some eight millenniums ago. These beads need to be properly identified and thoroughly studied. The line drawings of various artifacts (Fig.14, p.425) with a figure appearing like that of a dog's head seem to be highly imaginative. The sample of the perforated stone inspected under optical microscope (100 X) is said to have yielded conchoidal shaped irregular depression (Figs.15 and 16, p.426). However, these figures do not reveal such conchoidal fractures. Mottled appearance of the object (artefact no.28 inset figure) indicates that the object could be coarser grained sandstone but not a material that could yield conchoidal fracture such as crystalline or cryptocrystalline silica. It is unlikely that conchoidal fractures could be produced from such a specimen.
3. The rise and fall of sea level in the Late Pleistocene and early Holocene has been discussed by Nigam and Hashimi (2002) in response to Gupta's note (2002). A fall of ~40 m in ~9500 BP that stabilized for around 1000 years has been suggested by Nigam and Hashimi (*op. cit.*) which coincides with the likely existence of the present discovery.

S. Kathiroli, S. Badrinarayanan, D. Venkata Rao, S.N. Rajaguru, K.M. Sivakholundu and B. Sasisekaran, National Institute of Ocean Technology, Chennai - 600 031, reply:

We are thankful to Prof. R.V. Karanth for his valuable comments on our paper (Kathiroli et al. 2002). The clarifications sought by Prof. Karanth are as follows:

1. Correlation of E-W alignments of palaeochannels of the Tapi on the eastern side of the Gulf of Cambay with that of Satrunji river on the west is at present hypothetical and future work on sediment cores, which are being systematically collected all along the possible alignment will throw additional light on the validity or otherwise of our hypothesis.
2. The broken barrel shaped bead on chert is one of the most interesting finds from our collection so far. Making beads from hard stones like chert certainly requires an appropriate technology, especially a method of drilling. The earliest report of double tipped diamond drills used on hardstones is from Hazar-ar-Rayhani, Yemen, dated to around seventh to fifth century BC (Peter Francis, 2002). In the Indian sub-continent beads on chert/chalcedony materials were first produced at Mergarh (now in Pakistan) around 6000 BC (Jarrige et al. 1995).

   In the light of evidence in Yemen and Mergarh and our discovery of a broken bead on chert, dated to around 6000 BC, a new line of research in bead technology and trade is warranted.

   We are aware of difficulties involved in interpreting perforations in sandstone. It is not easy to separate anthropogenic factors from natural ones in this case. Our attempt in resolving this problem by using optical microscopy with 100x magnification is only a beginning; further detailed investigations are in progress.

3. We are thankful to Prof. Karanth for bringing to our notice a very useful note on sea level changes and human settlements in the Gulf of Cambay by Drs. Nigam and Hashimi of NIO, Goa. Their expert views on sea level changes and archaeological settlements around 8000 BP will be taken into consideration in our future work in the Gulf of Cambay.

Lastly, our paper is a preliminary report on the archaeological finds discovered in the sea bed at a depth of 30-40 m below the present sea level. We have succeeded in generating interest of archaeologists, geologists, geochronologists and other natural scientists in India and abroad. Our next two years of multi-disciplinary and multi-institutional studies are expected to contribute in a significant way to marine archaeological research in India and the antiquity of our pre-historical records.

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BOOK REVIEW

SGAT BULLETIN, (v.3, no.1, 2001), Journal of the Society of Geoscientists and Allied Technologists, C 73 HIG, Baramunda Housing Colony, Bhubaneswar - 751 003

The Society of Geoscientists and Allied Technologists, Bhubaneswar, has come into existence since 1980 and for the last three years has been publishing the journal entitled 'SGAT Bulletin'. Research papers, review articles, short communications, announcements and letters to the editor are invited on topics of geosciences, mineral exploration, mining, materials science, metallurgy, mineral industry and trade, environment, education, R&D, legislation and infrastructure related to mining, mineral policy and mineral development planning.

The journal under review is the third volume of the Society (v.3, no.1, 2001). It is a volume containing about 84 pages and encompasses four research papers, two review papers taking up about 60 pages and 25 pages are devoted to several news items. Though the journal has been started with good intentions of quick publication of the research findings in the related fields, it needs to be improved to a great extent. The cover page outlay and colour could have been more subtle to attract the readers.

The reviewer has not gone into the scientific contents of the journal as it is beyond the purview, yet there is a feeling that the scientific standard of papers has to be evaluated and scrutinized more stringently, so that material of high standard only is published. It is felt that there is no need of tabulated columns of analysis and other raw data and only the highlights can be listed and summarised. The quality of the illustrations like geological maps, photographs etc. although appear acceptable, yet there is ample scope for improvement. A lower font size, a double column printing and avoiding unnecessary data sheets will definitely